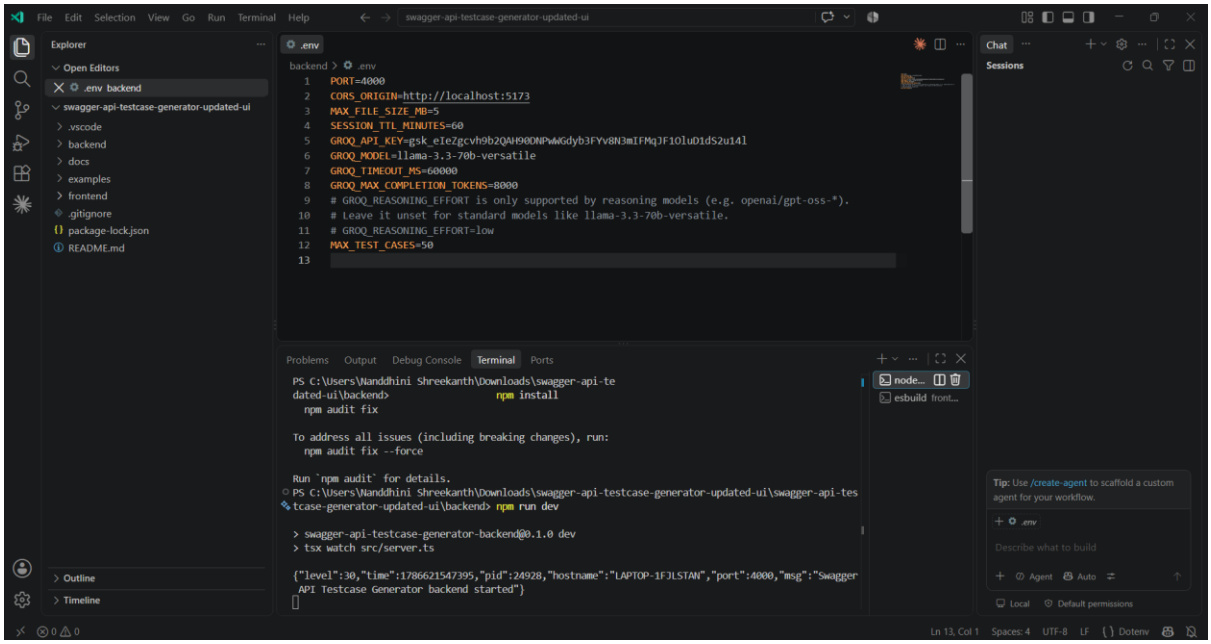
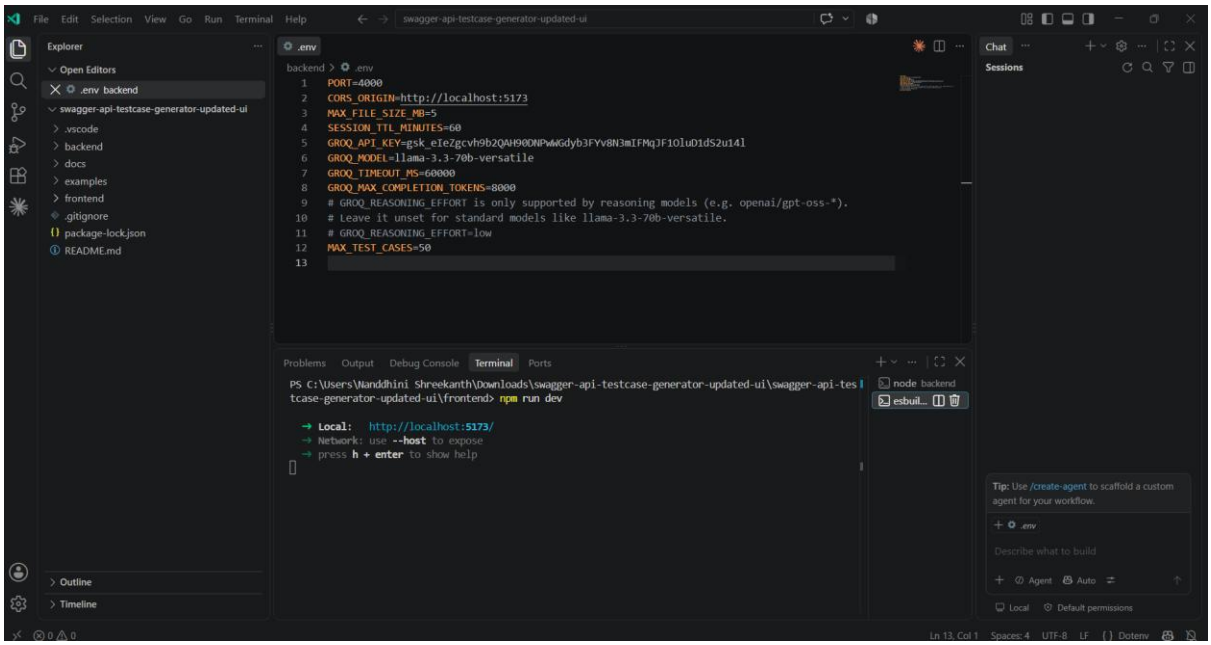
1. **Generate Reliable API Test Cases from Swagger**

Restful-booker-openapi.yaml

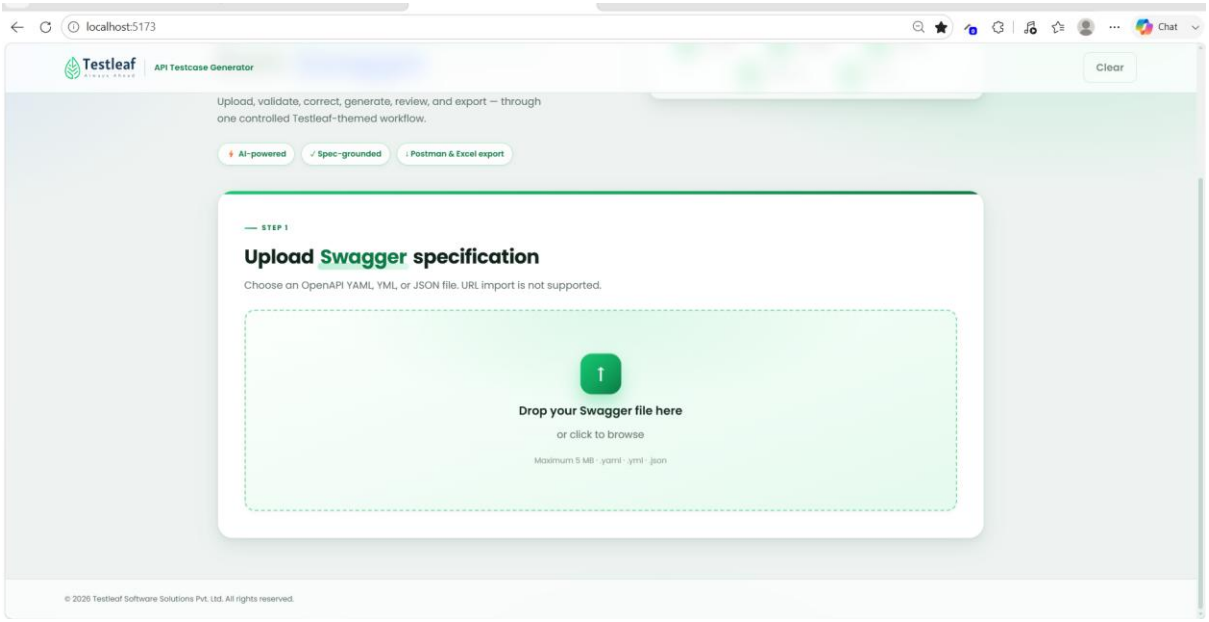
Run the backend with npm run dev



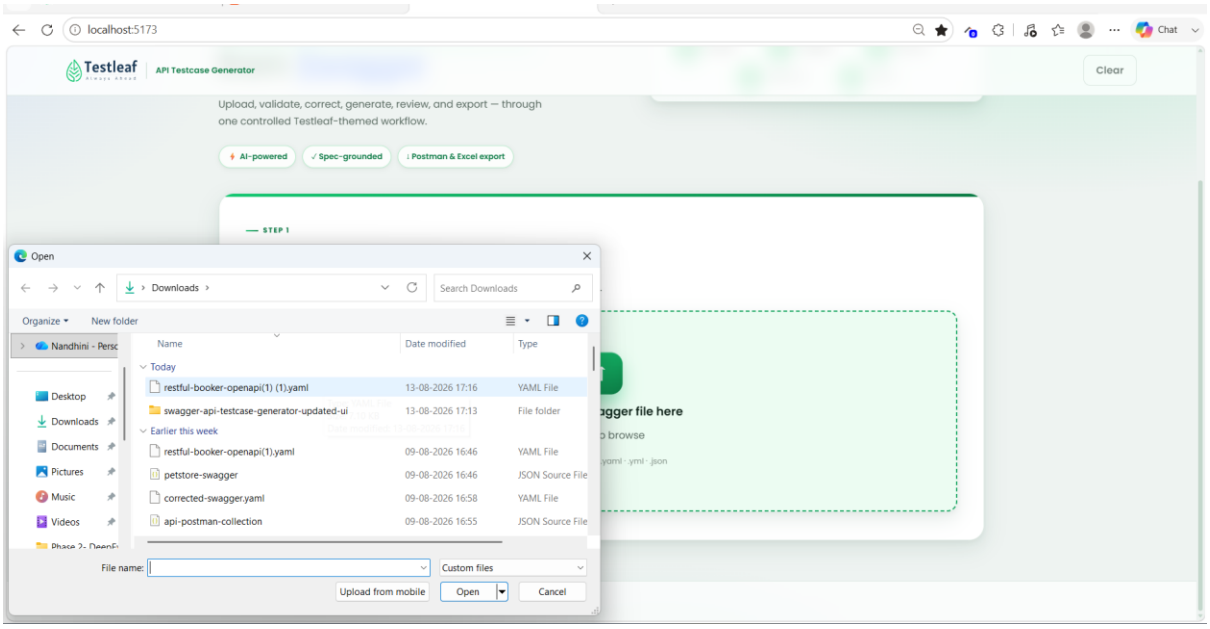
Run the front end with npm run dev



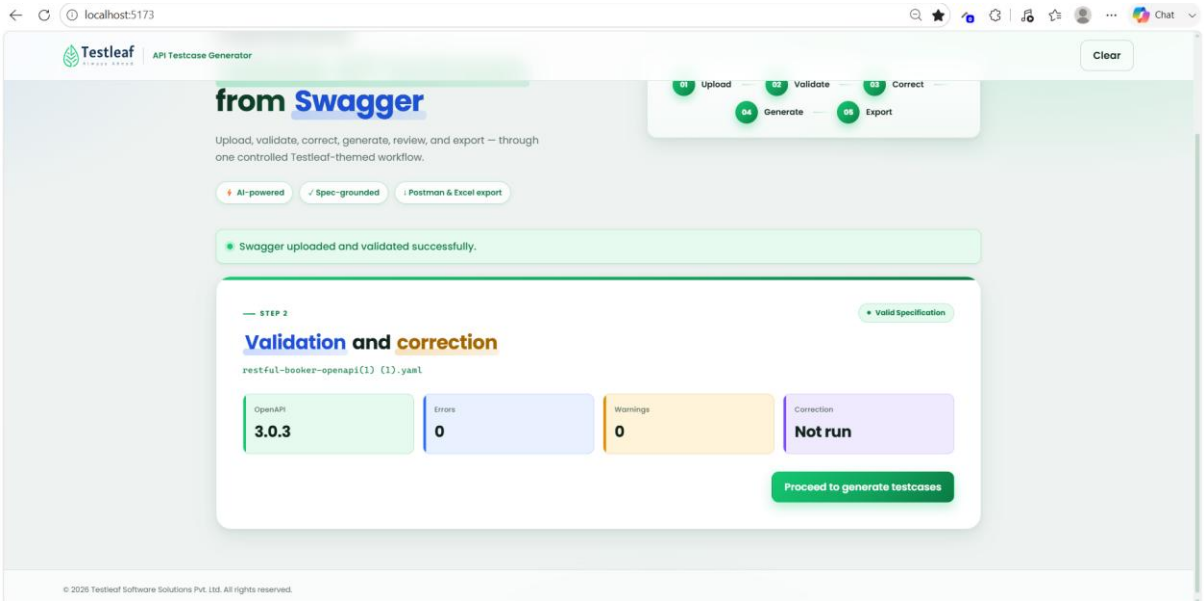
UI



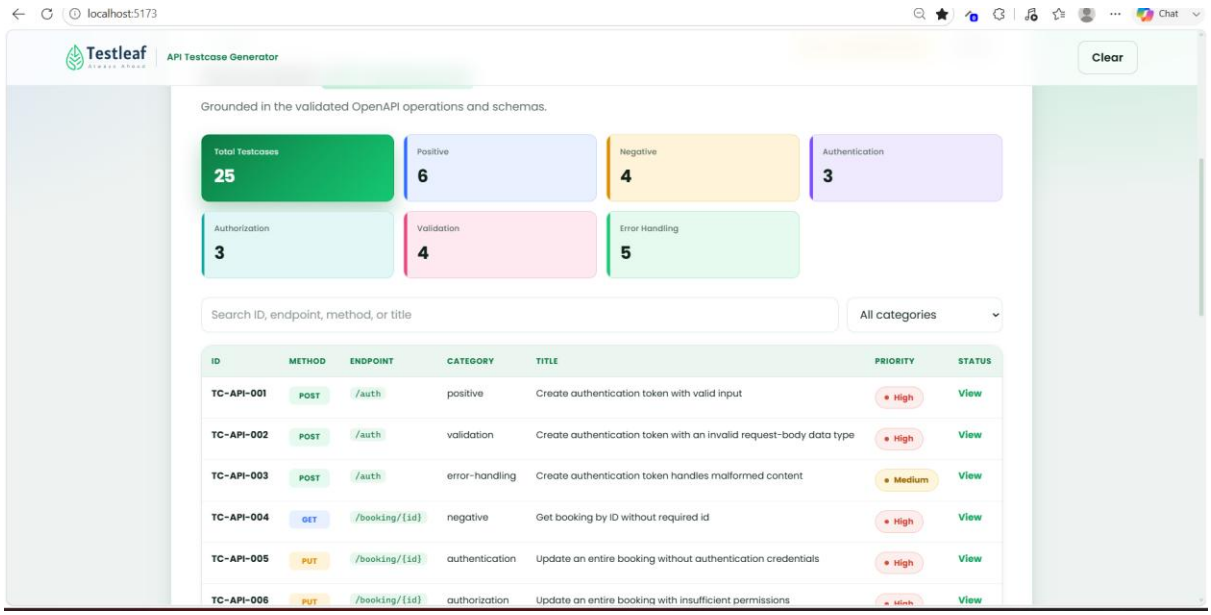
Uploading restful booker.yaml file



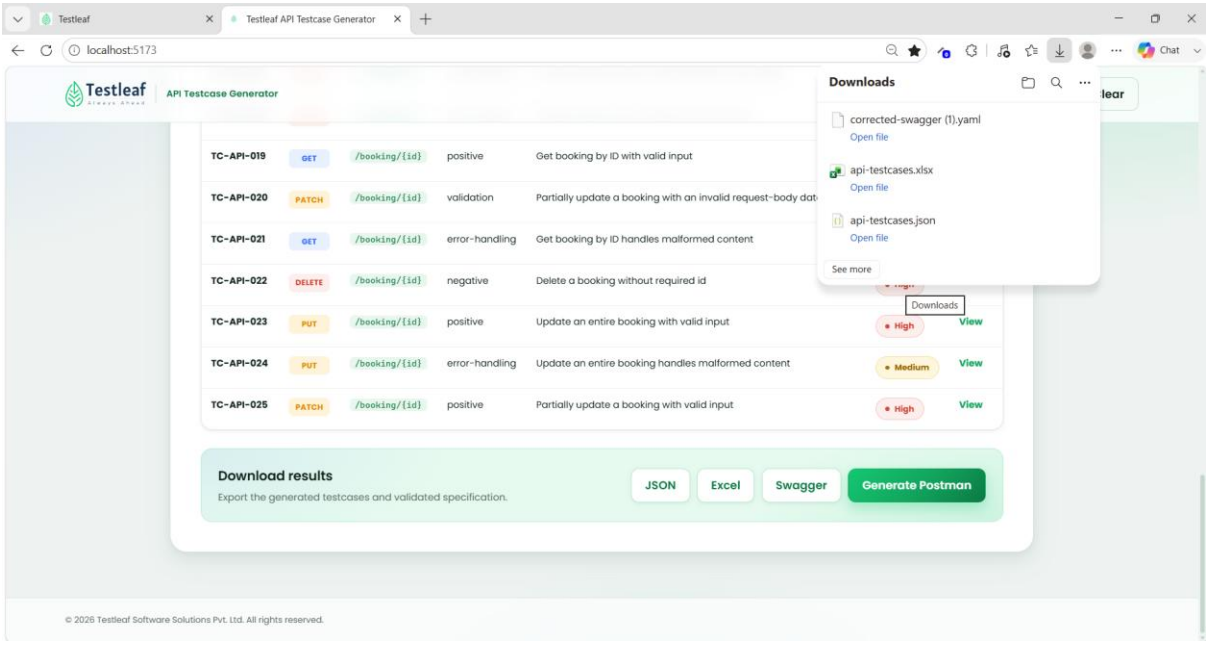
Uploaded successfully with no errors



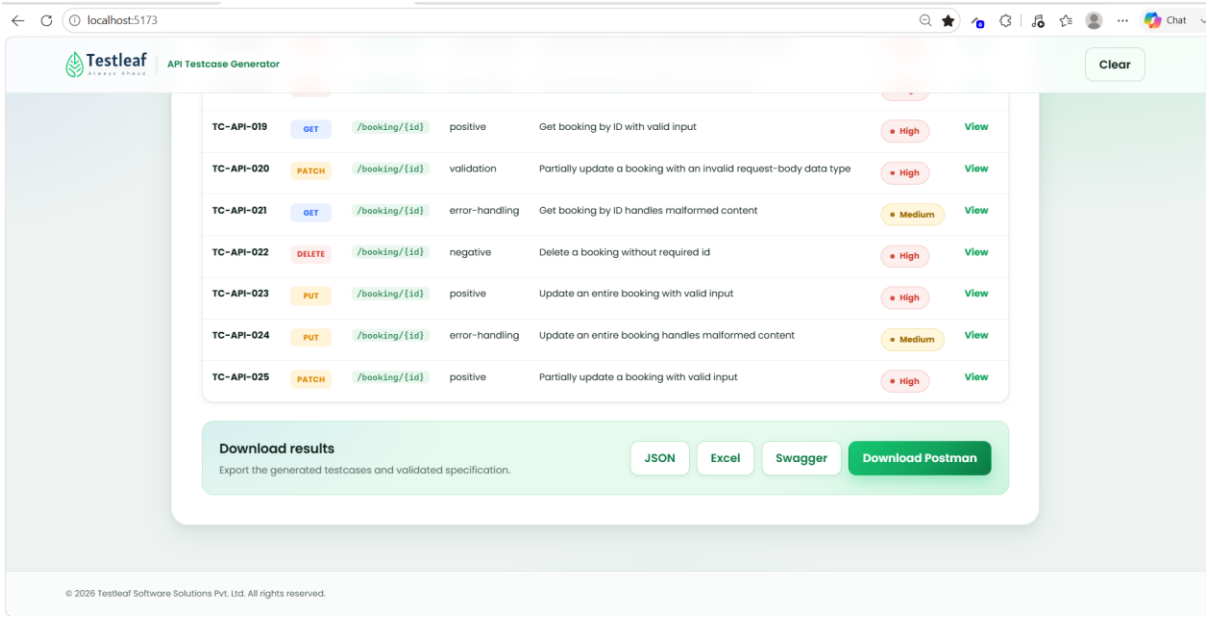
Click Proceed to generate testcases



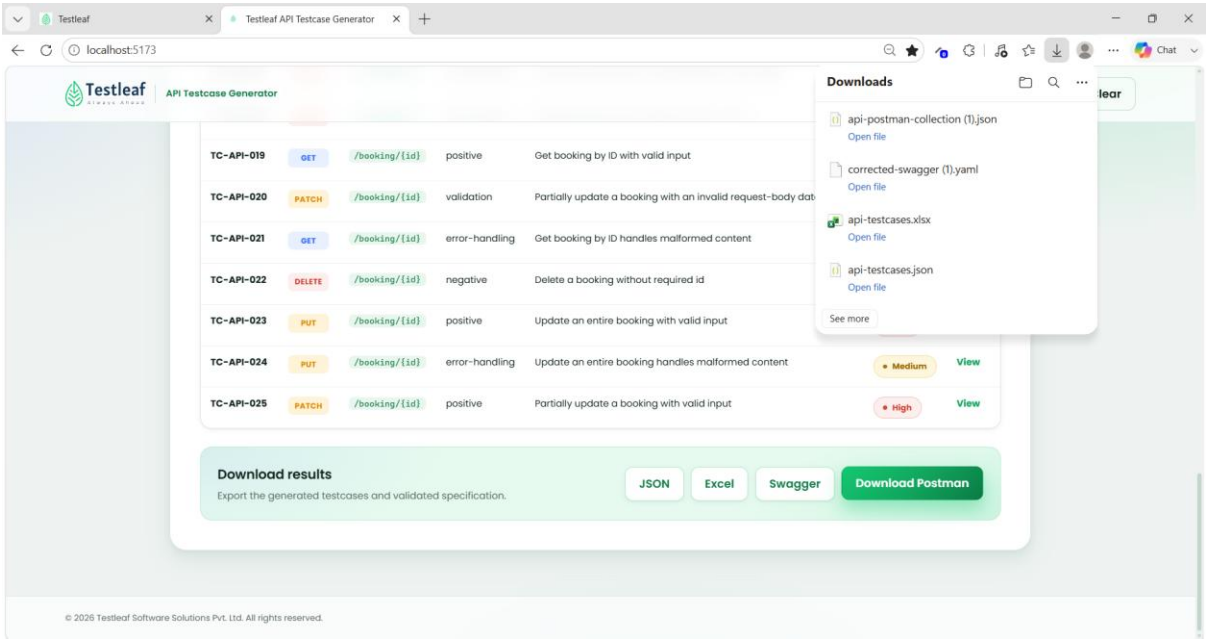
Export the generated test cases through Json/Excel/Swagger



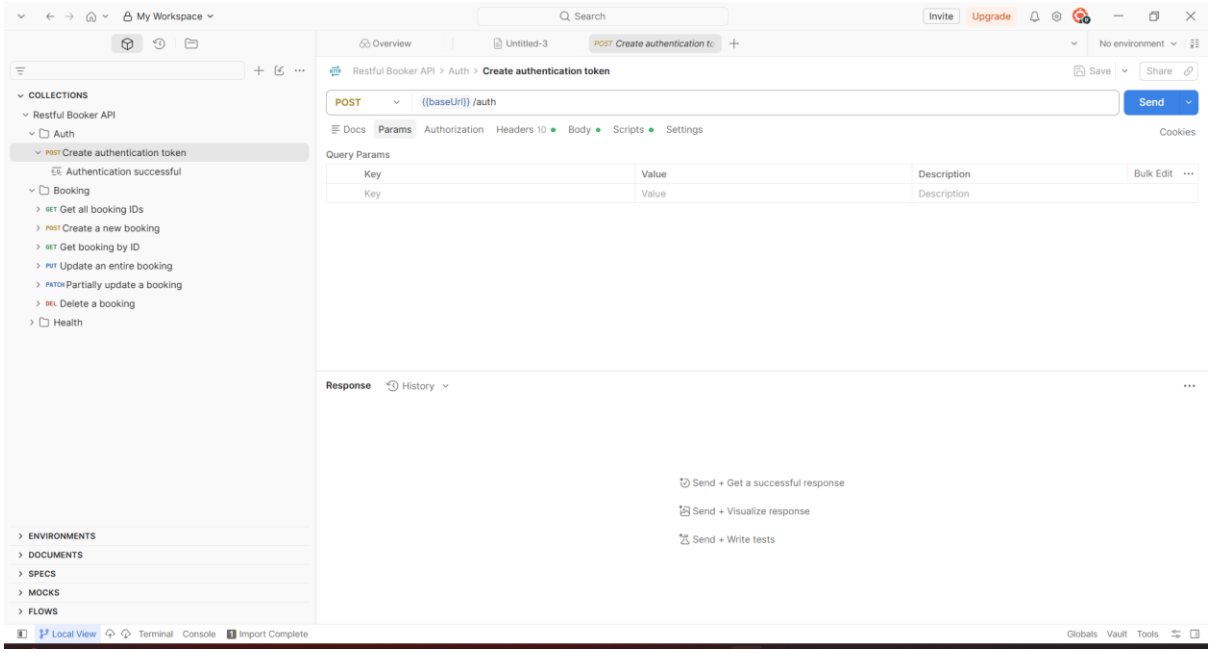
Generate Postman Collection is clicked, it will ask for download postman



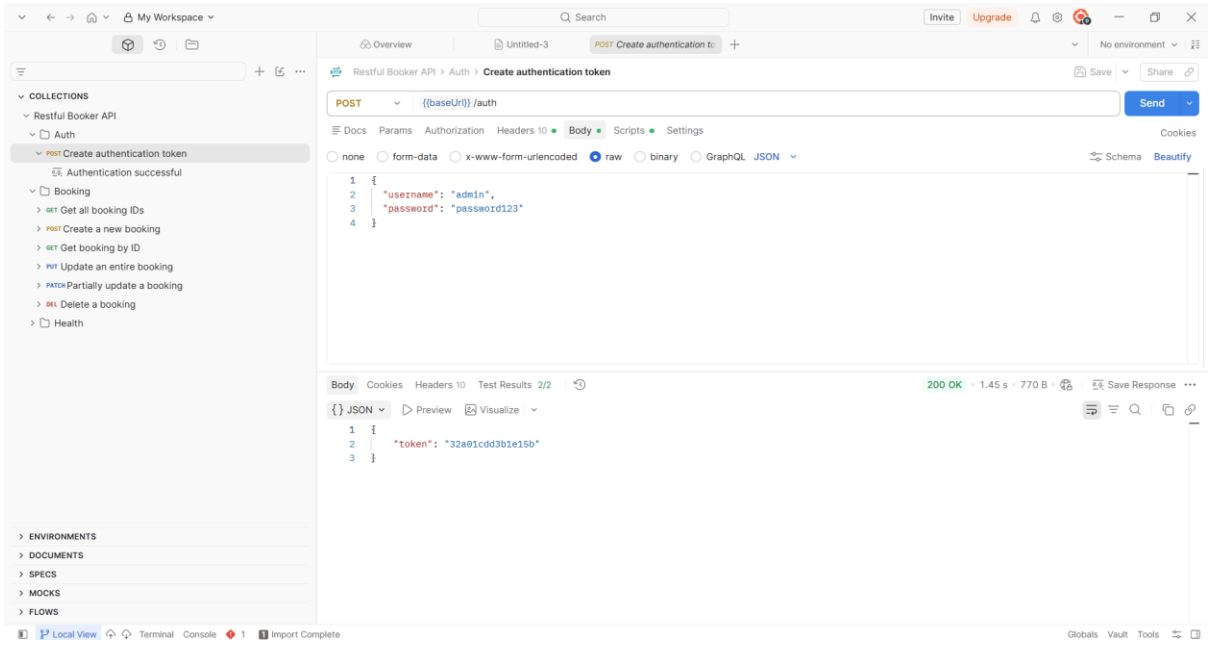
Click Download Postman



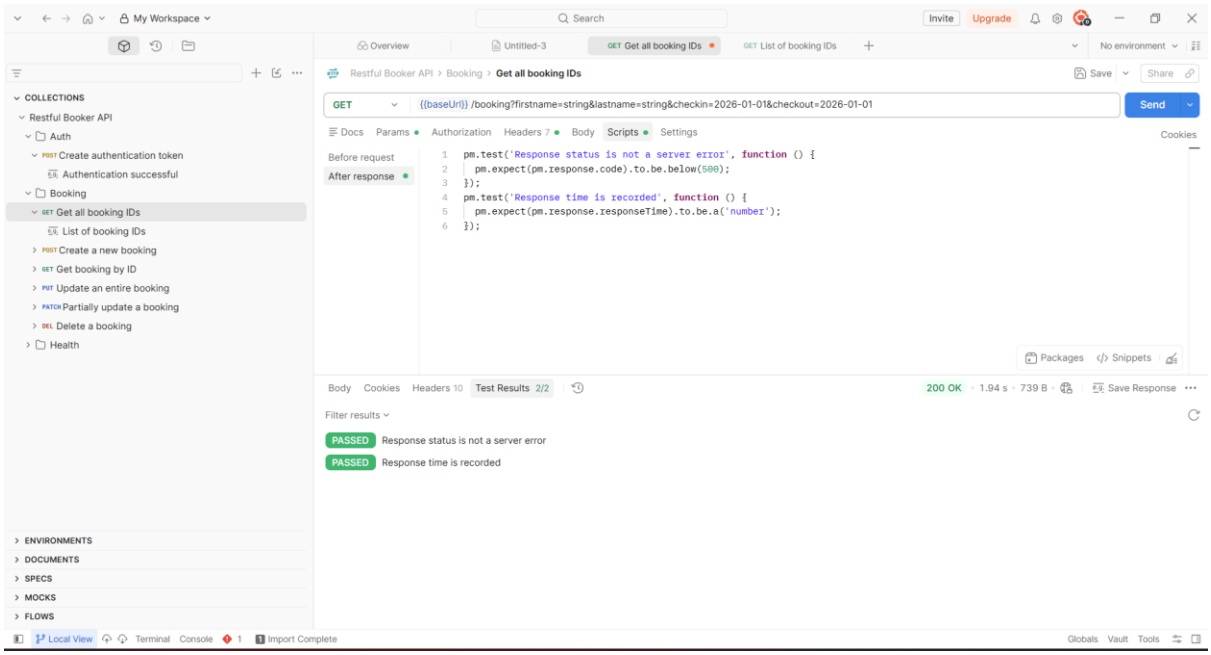
Open the downloaded Postman file in Postman



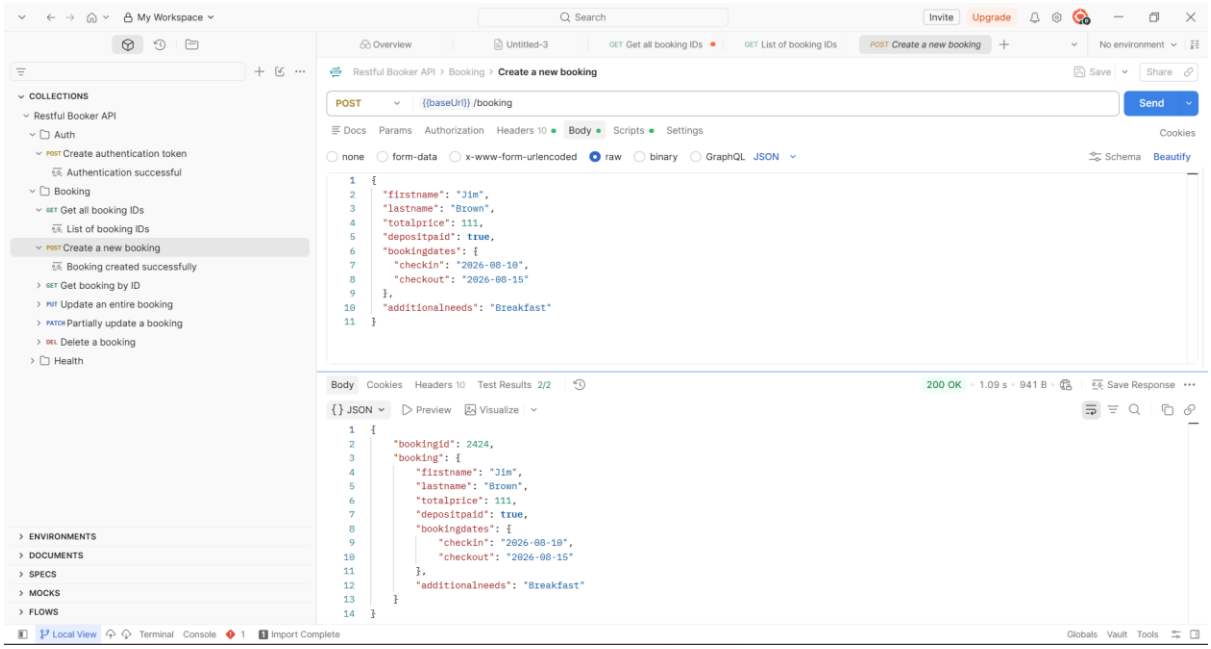
Verified Create Authentication token



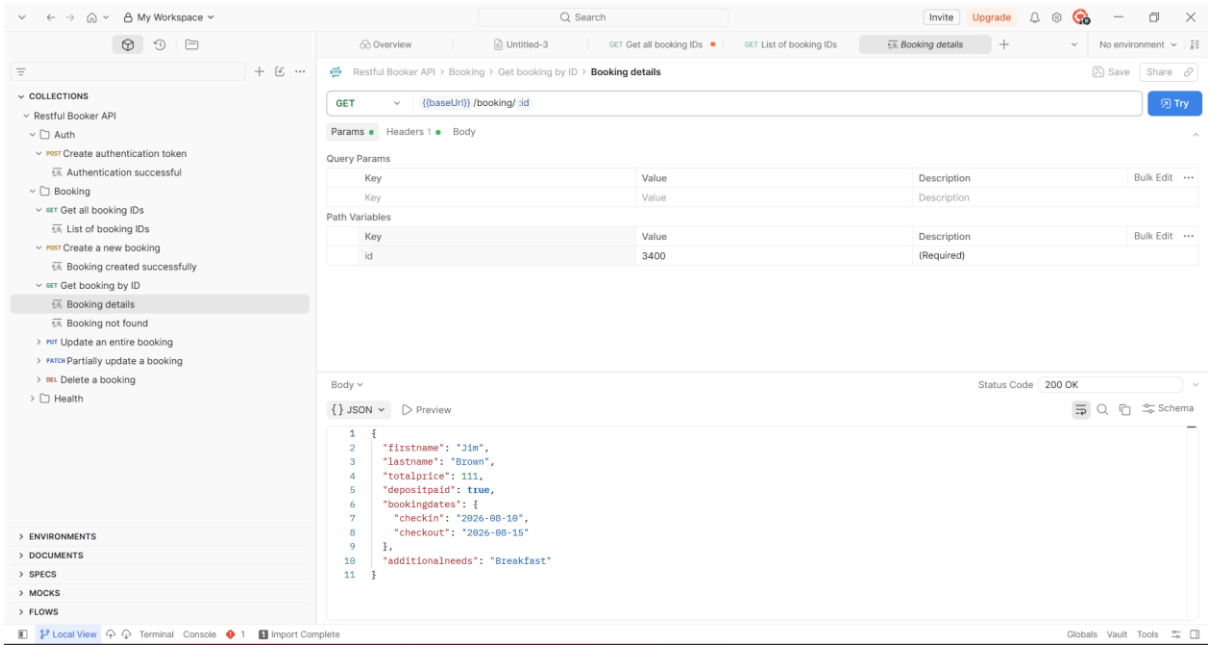
Verified Get All Booking IDs



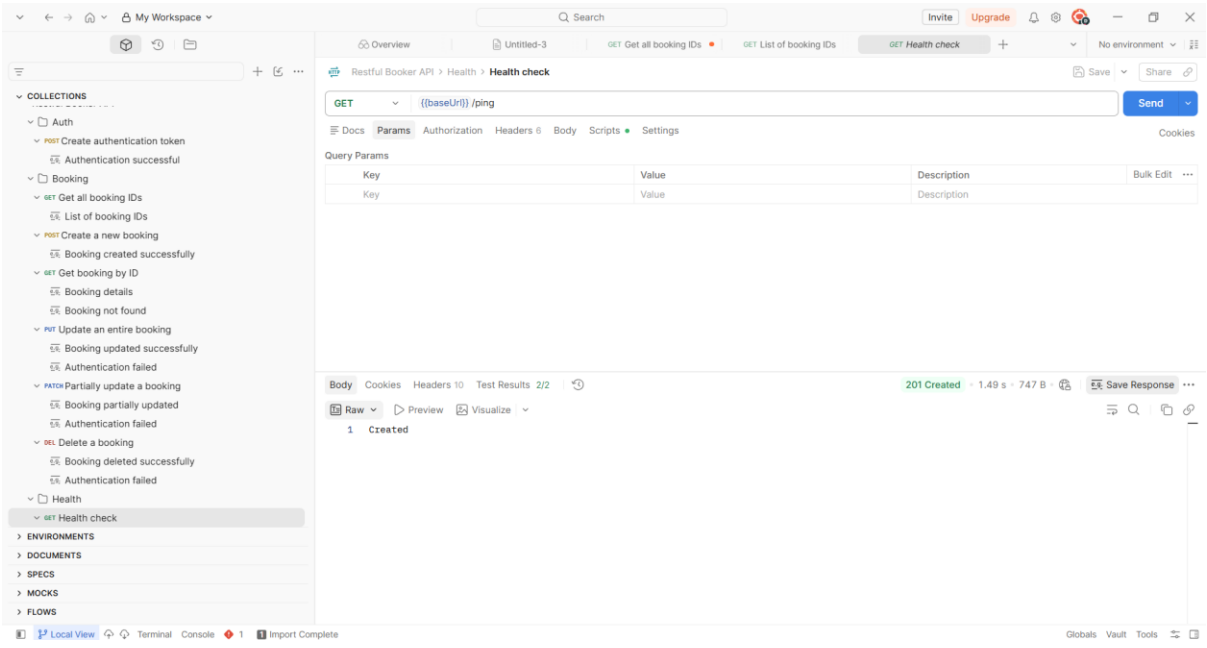
Create a new booking ID



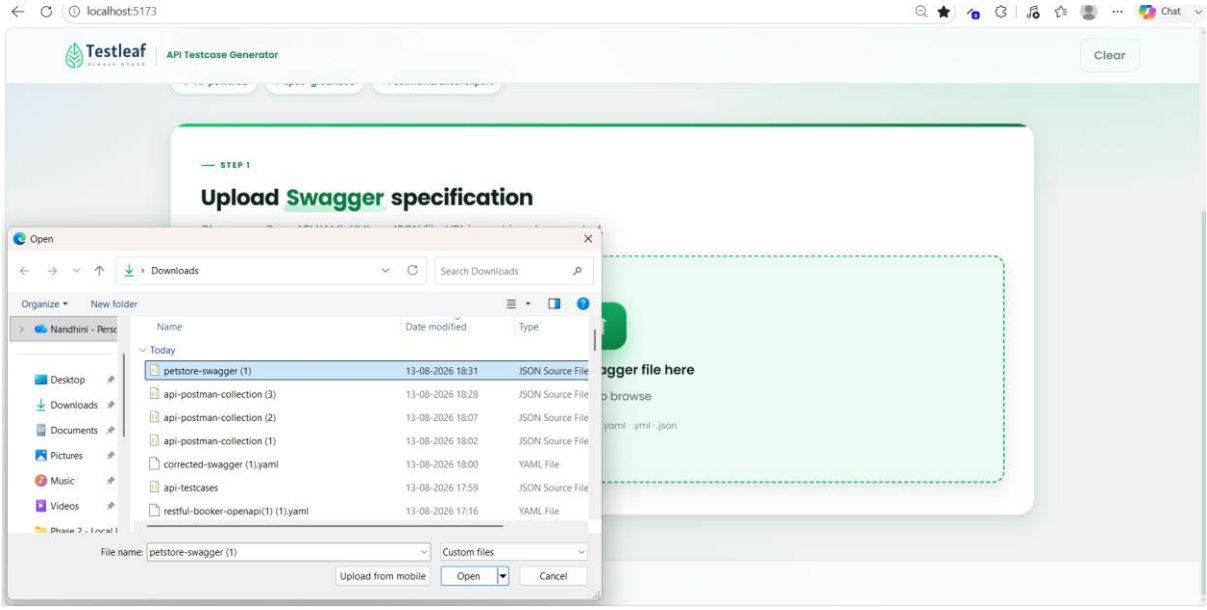
Get booking details



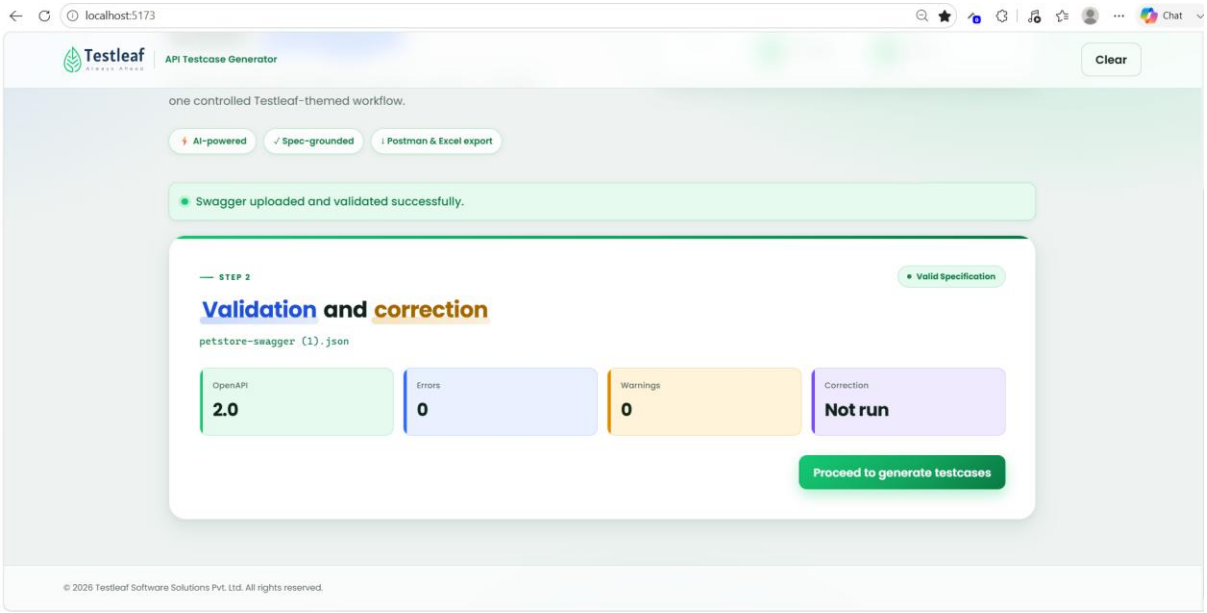
Get Health check



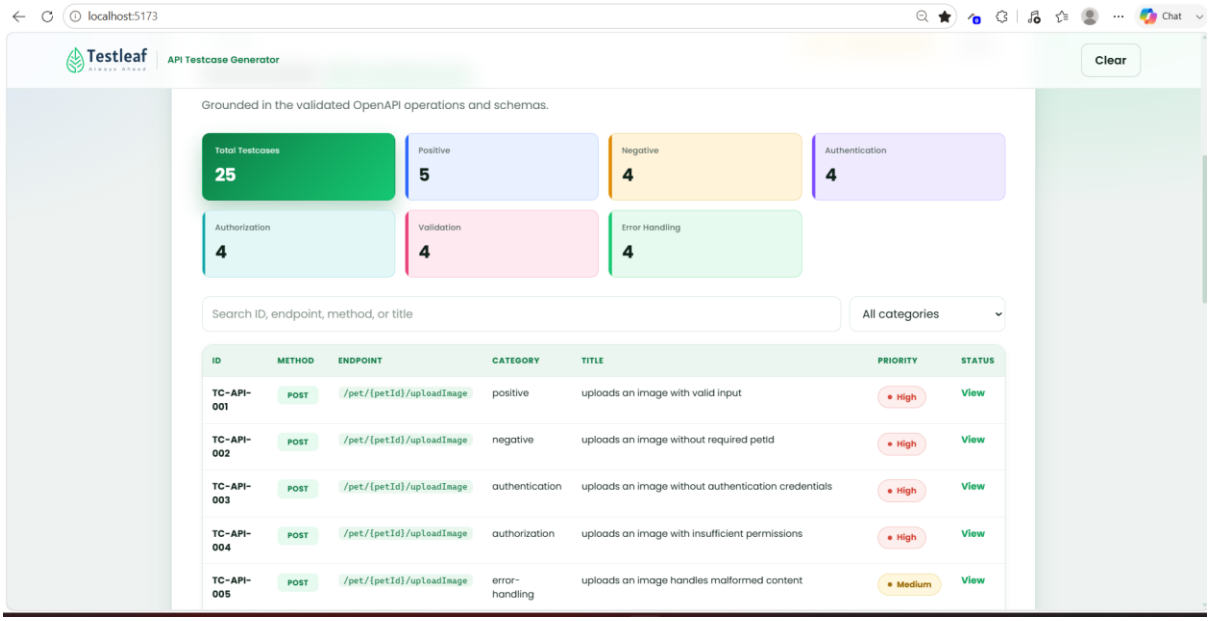
Upload .json file



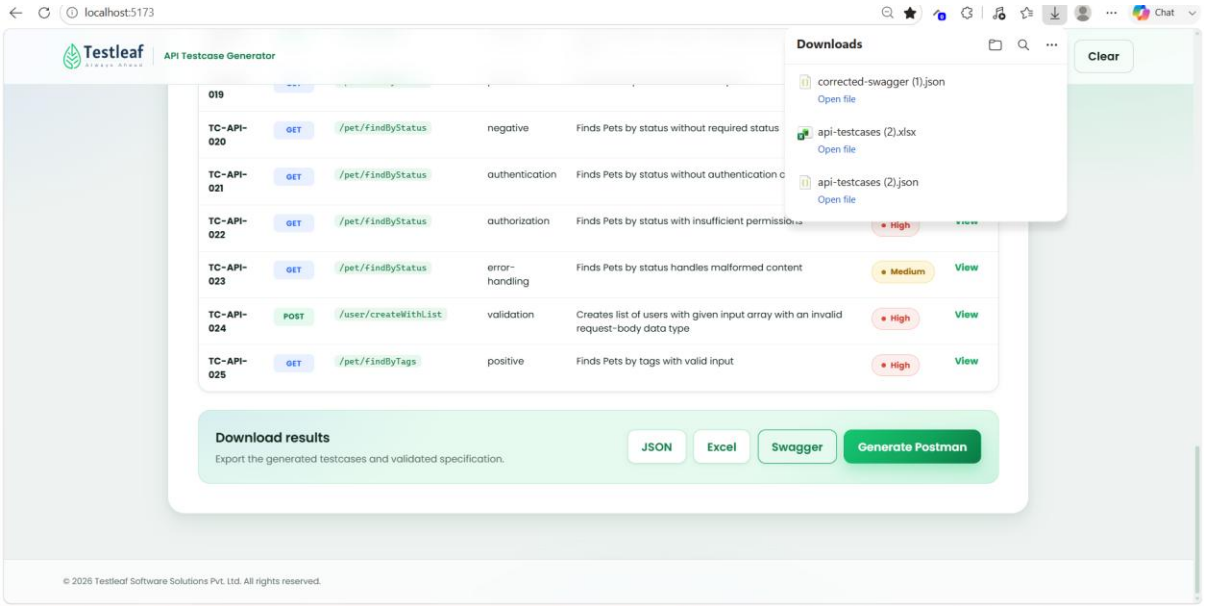
Processing is done with no errors



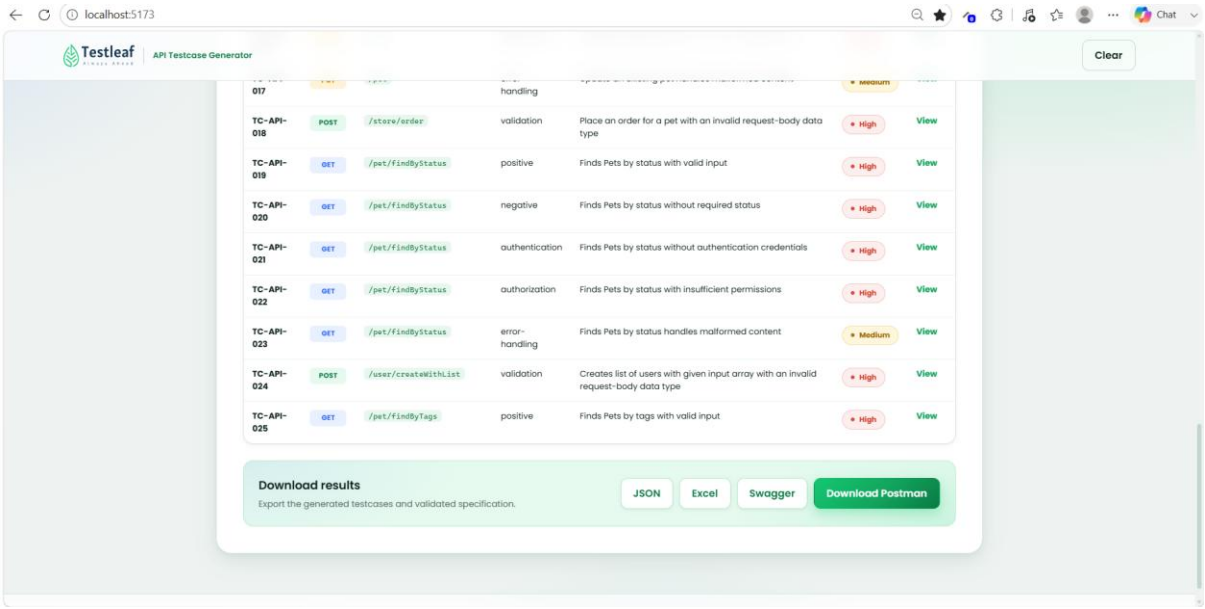
Click on Proceed to generate testcases



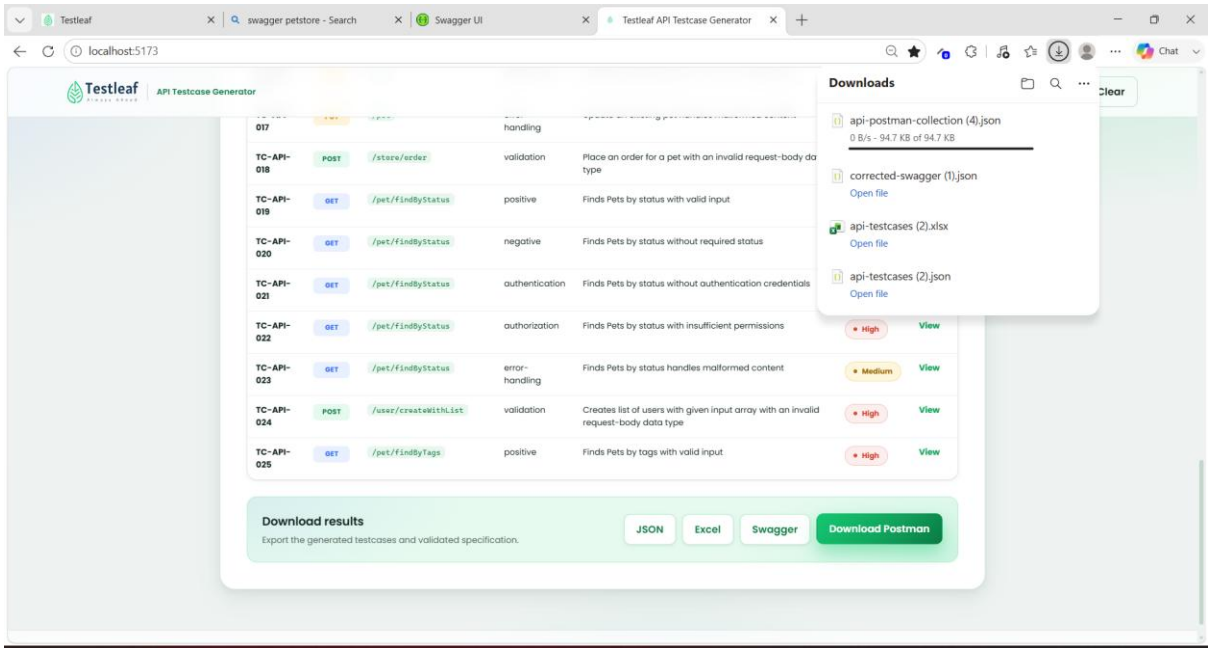
Export the generated test cases through Json/Excel/Swagger



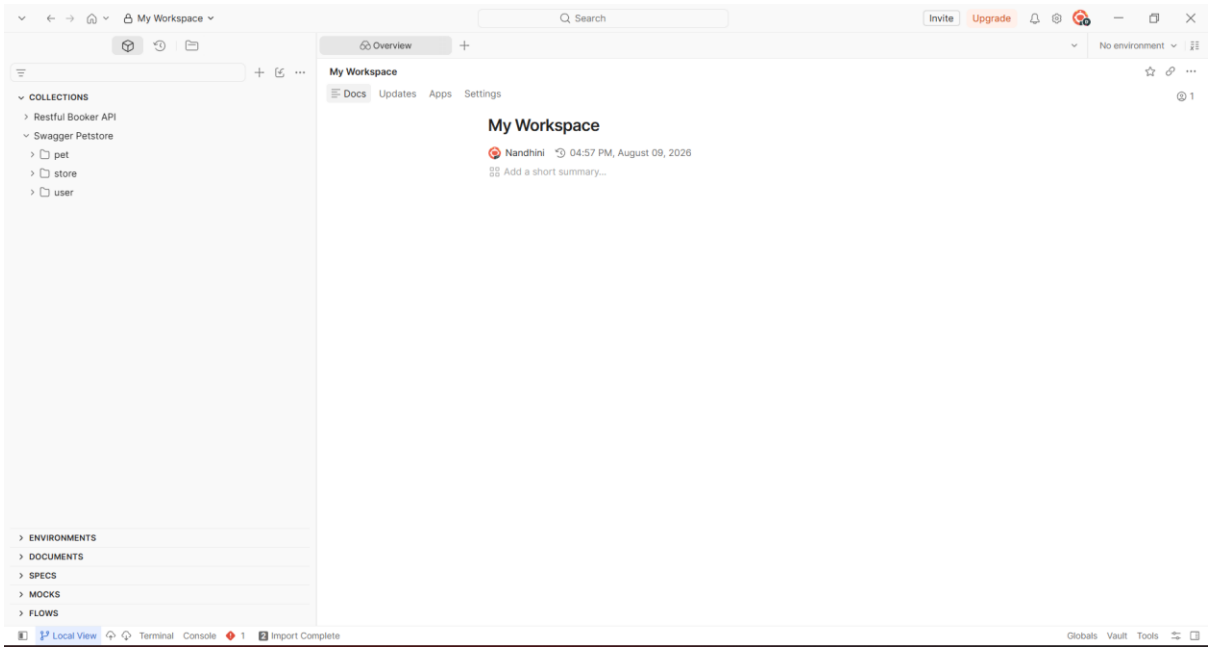
Generate Postman Collection is clicked, it will ask for download postman



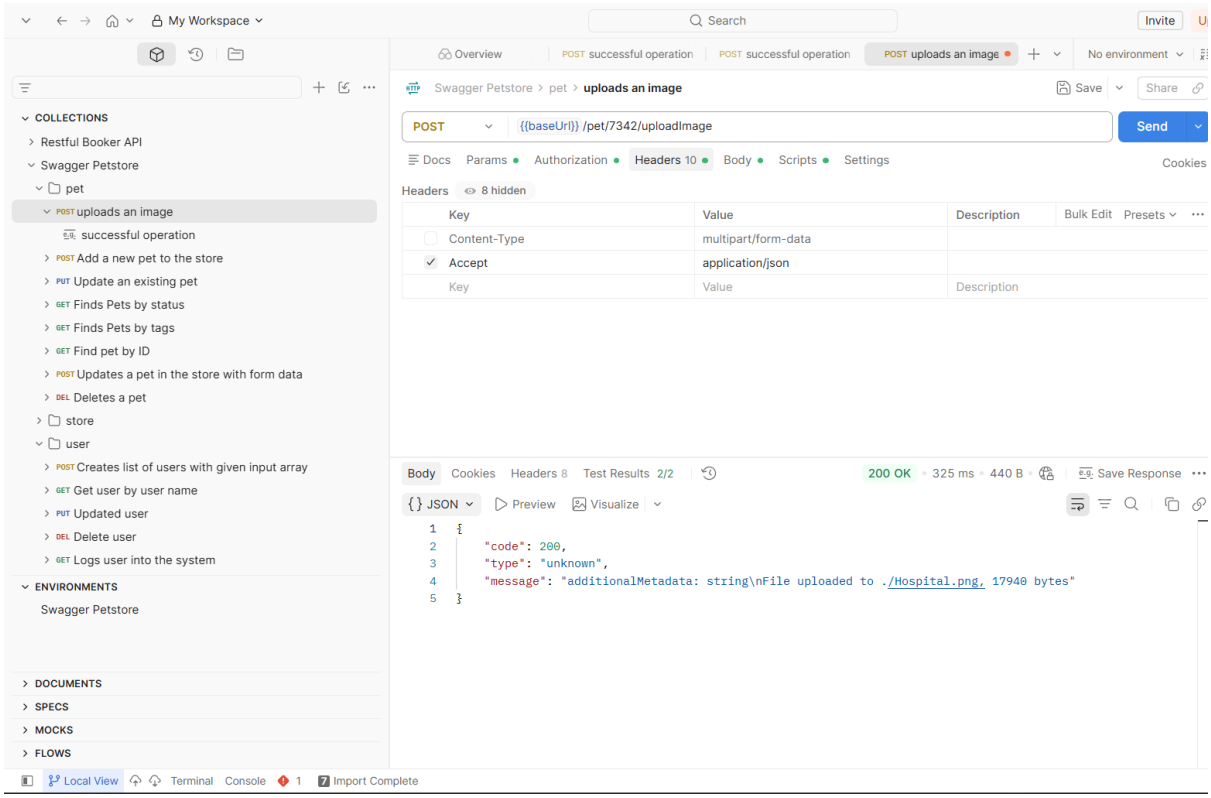
Click Download Postman



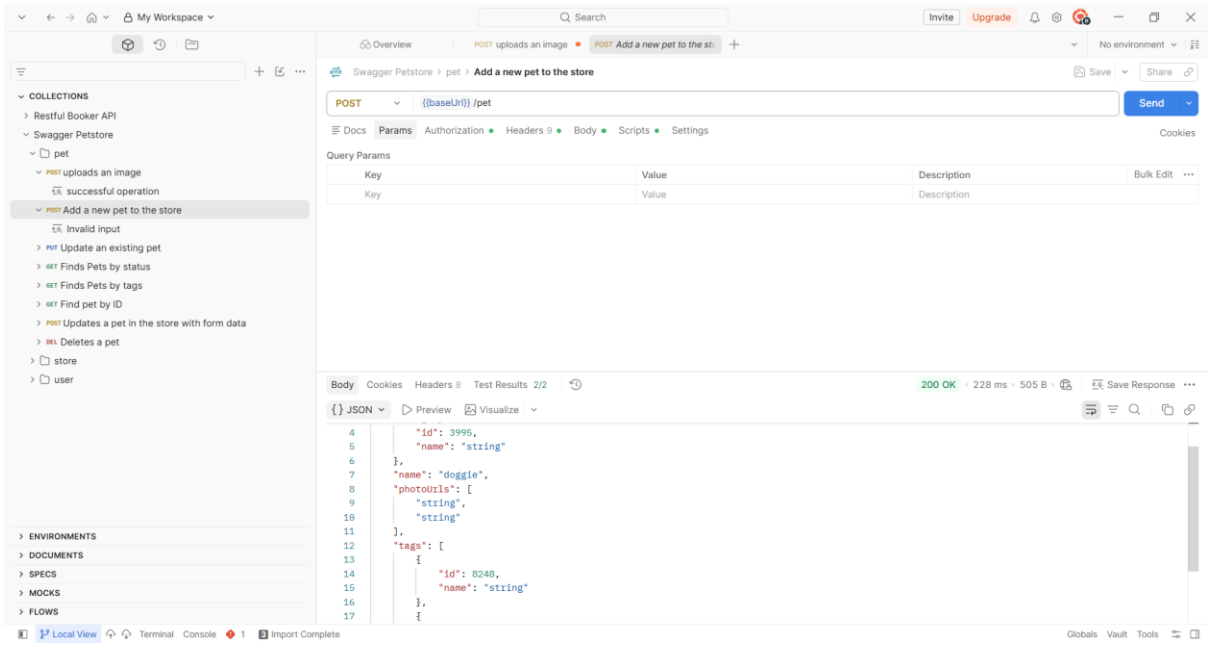
Open the downloaded Postman file in Postman



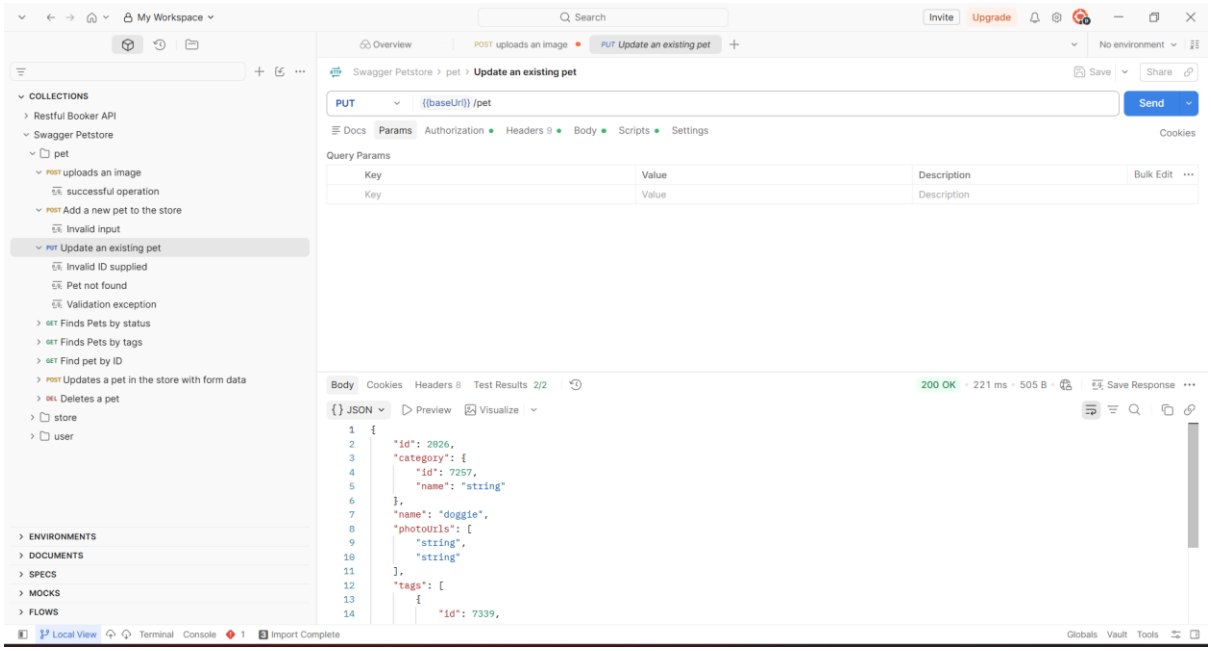
Upload image



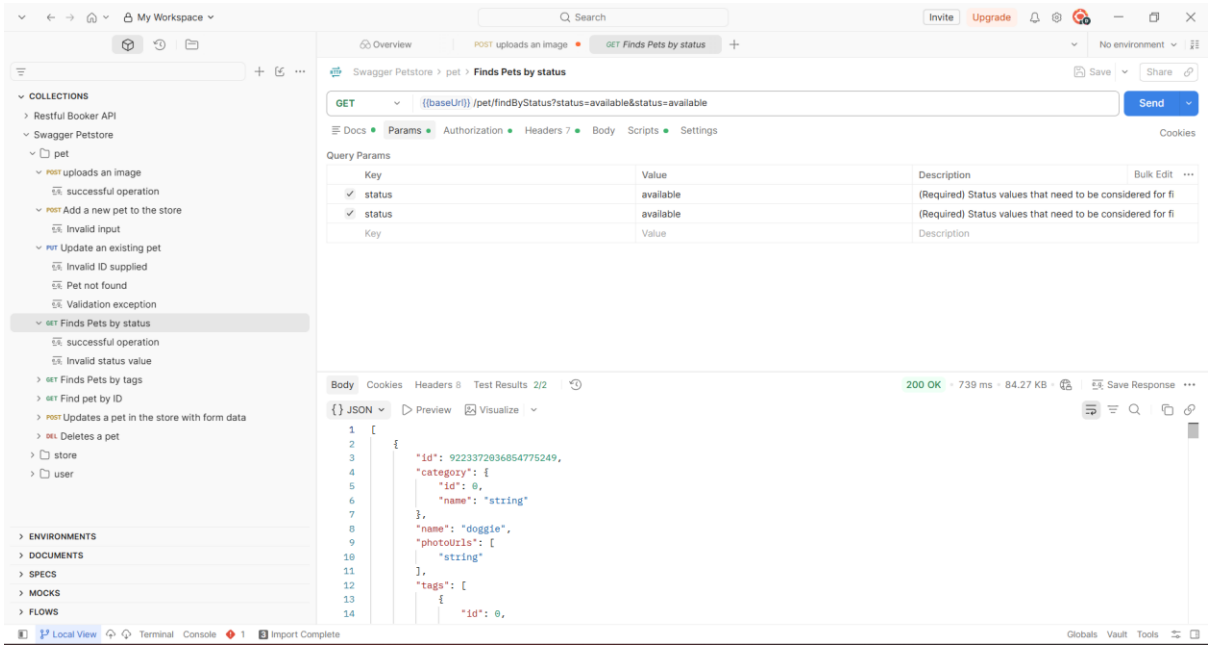
Add a new pet to the store



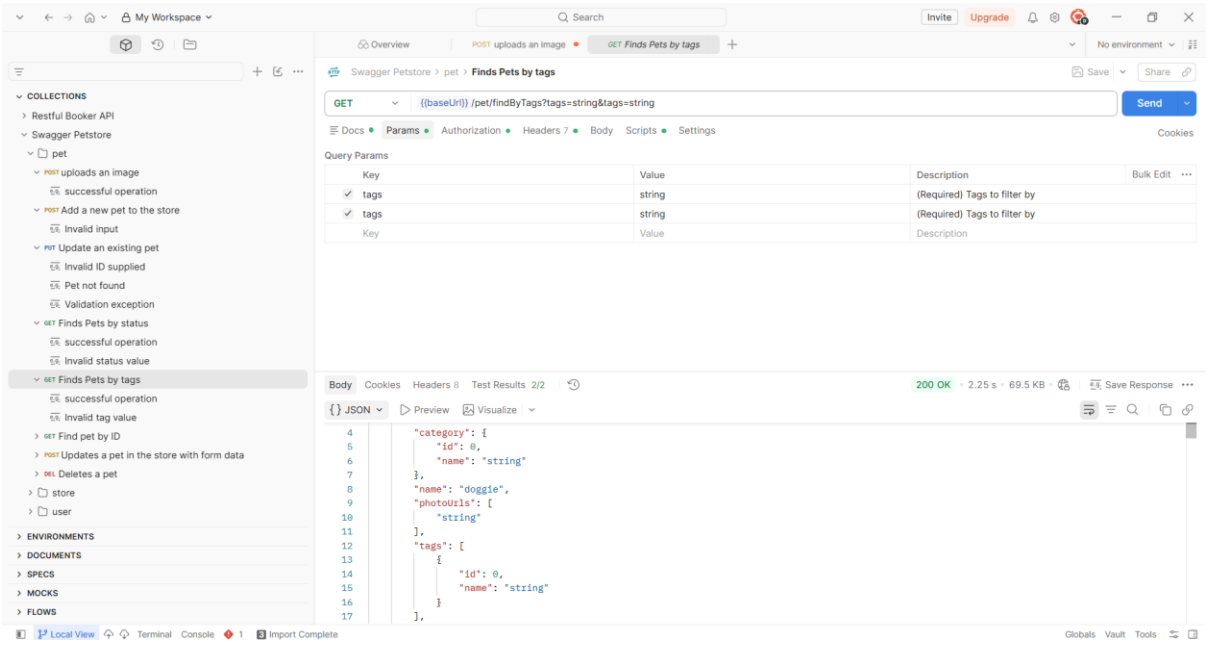
Update an existing pet



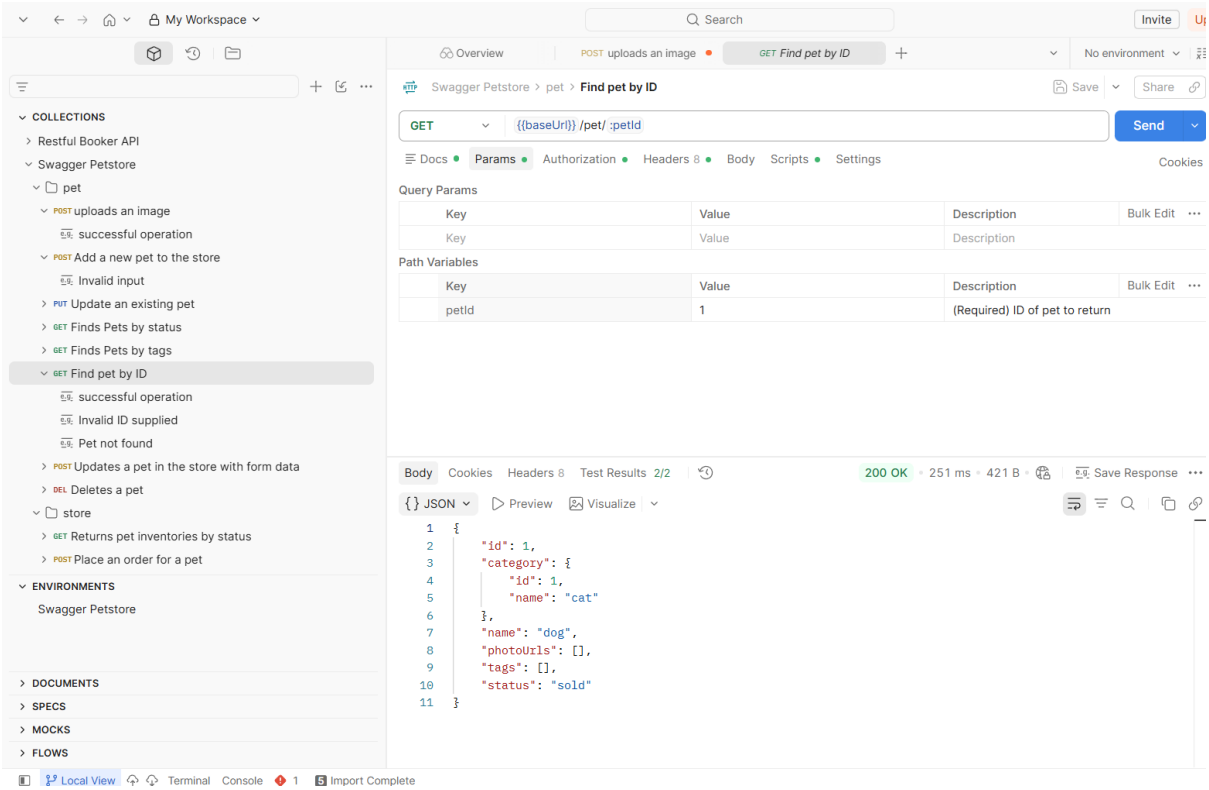
Find pet by status



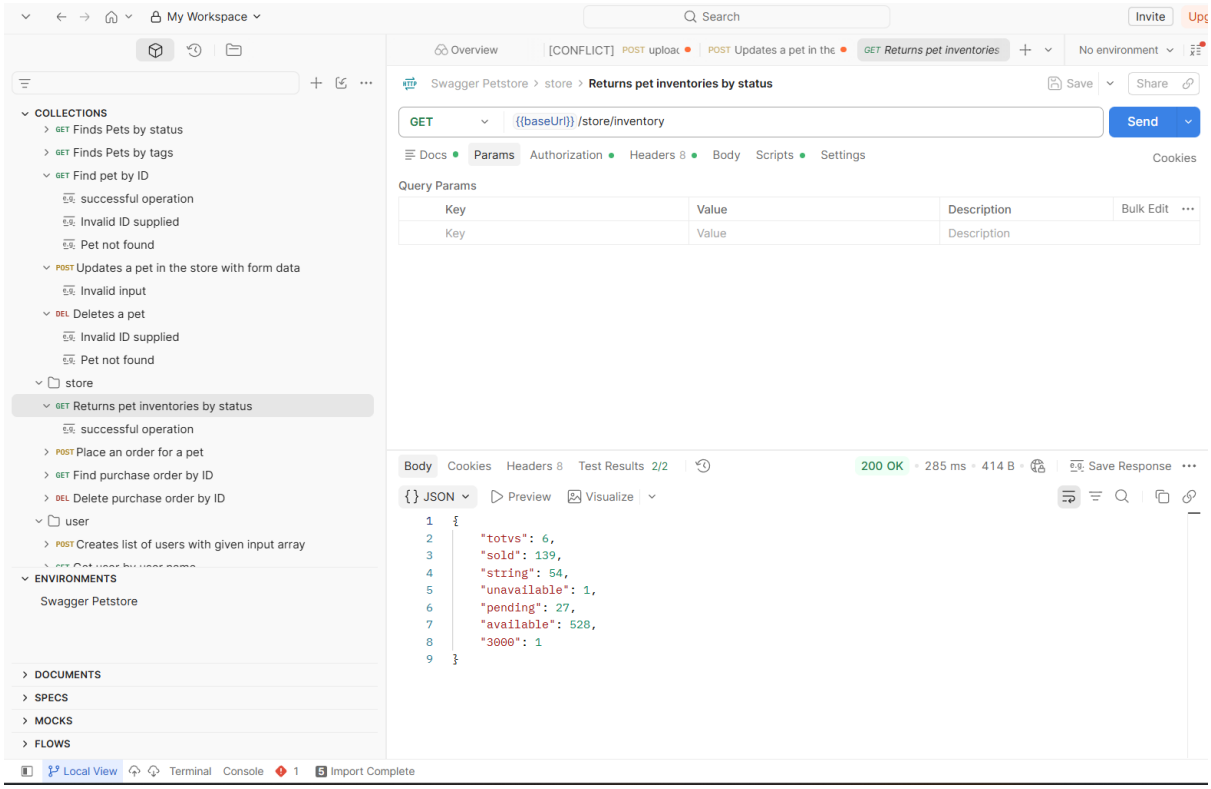
Find pet by tags



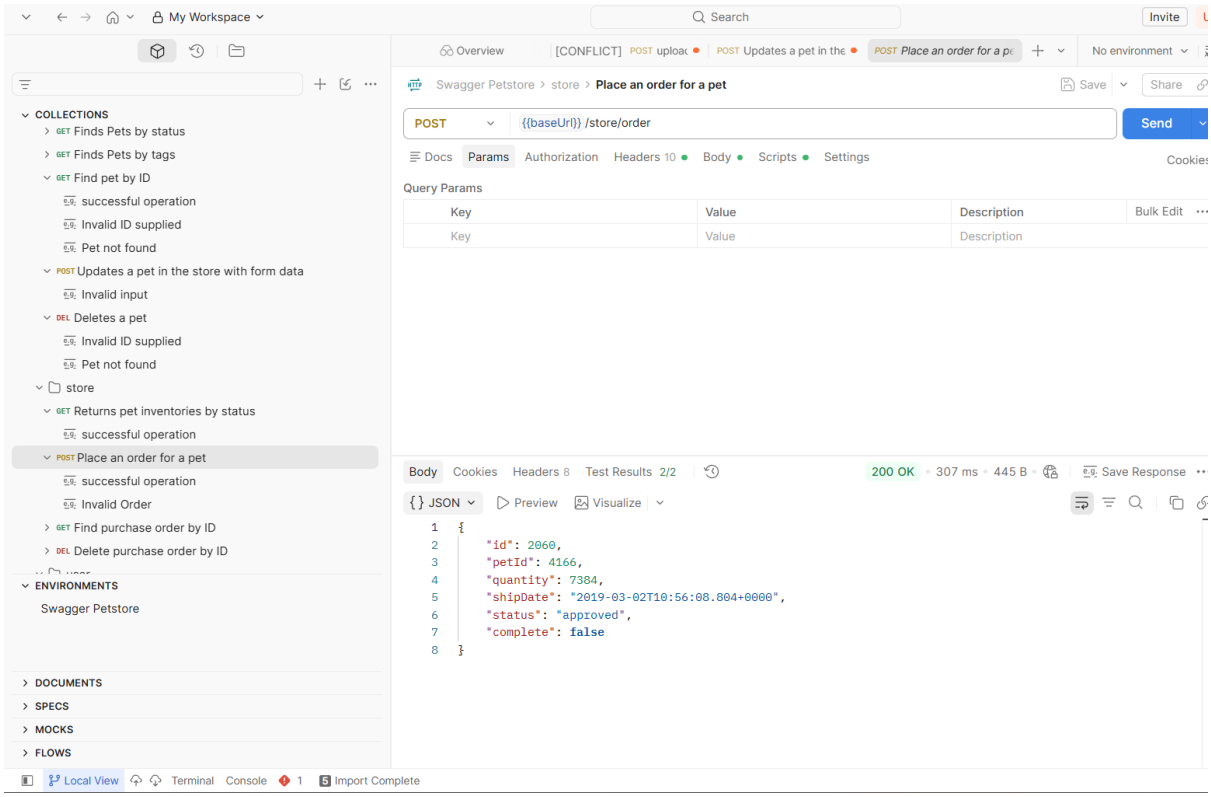
Find Pet by ID



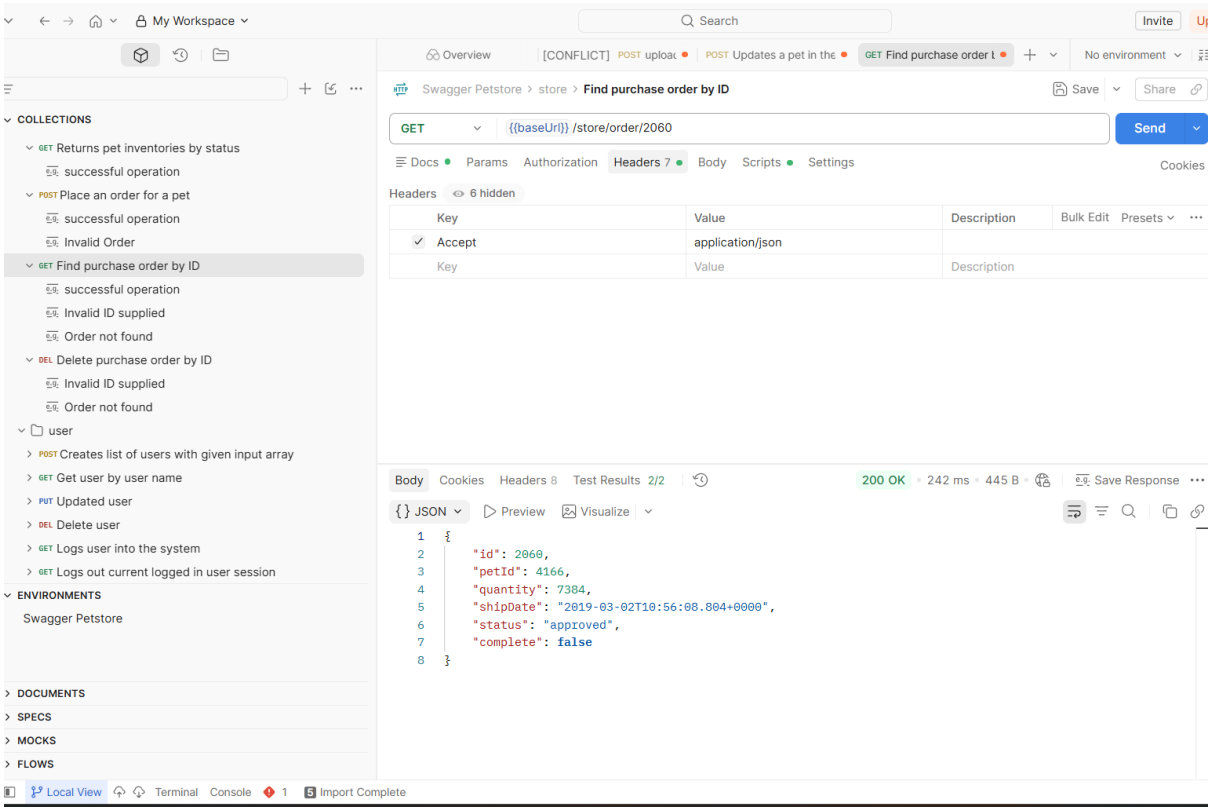
Returns pet inventories by status



Place an order for a pet



Find purchase by order ID



Gave the prompt as Modify the **existing API Testcase Generator application** shown in the current UI.

Objective

Add a **Swagger/OpenAPI URL input option** so users can enter a Swagger specification URL instead of uploading a Swagger file.

Example URL:
https://petstore.swagger.io/

UI Changes

In the existing **Step 1 – Upload Swagger specification** section:

1. Keep the existing file-upload functionality unchanged.
2. Add a clearly visible **"OR"** separator below the file upload area.
3. Add a textbox/input field with:
    - Label: **Swagger URL**
    - Placeholder: **Enter Swagger/OpenAPI specification URL**
    - Example value: https://petstore.swagger.io/

1. The UI should remain consistent with the existing Testleaf design:

    - Green theme
    - Existing fonts, spacing, cards, borders and buttons
    - Responsive layout
2. Add basic URL validation and show a user-friendly validation message for an invalid/empty URL.

URL Processing

When the user enters:

https://petstore.swagger.io/

the application should retrieve the Swagger/OpenAPI specification from that URL.

Important:

- Do NOT assume that the URL itself is always the raw JSON/YAML specification.
- Handle Swagger UI URLs where possible.
- Detect the actual OpenAPI/Swagger specification URL from the Swagger UI configuration.
- Support OpenAPI JSON/YAML and Swagger 2.0 specifications.
- If the specification cannot be retrieved, display a clear error message.

Backend

If the existing application has a Node.js/Express backend, create an API endpoint such as:

POST /api/swagger/import-url

Request:

```
{
  "url": "https://petstore.swagger.io/"
}
```

The backend should:

1. Validate the URL.
2. Fetch the Swagger/OpenAPI specification.
3. Parse JSON/YAML as appropriate.
4. Validate that it is a valid Swagger/OpenAPI specification.
5. Return the parsed specification to the frontend.
6. Reuse the application's existing test-case-generation logic rather than creating a separate generation implementation.

Test Case Generation

After successfully retrieving the Swagger specification:

1. Extract all available API endpoints.
2. Extract HTTP methods.
3. Extract path parameters.
4. Extract query parameters.
5. Extract request headers.
6. Extract request body/schema.
7. Extract response codes and response schemas.
8. Generate test cases using the application's existing test-case-generation workflow.

Generate positive, negative and boundary test scenarios where applicable.

For example, for the Petstore API, generate test cases for endpoints such as:

- GET
- POST
- PUT
- DELETE

Do not hard-code Petstore endpoints. The implementation must work with any valid Swagger/OpenAPI specification.

Loading and Error Handling

While the Swagger URL is being processed:

- Disable the Generate button.
- Display a loading indicator such as **"Fetching Swagger specification..."**
- Prevent duplicate requests.

Handle:

- Invalid URL
- URL not reachable
- Timeout
- Invalid Swagger/OpenAPI specification
- Unsupported specification format
- CORS/network errors
- Empty specification
- Backend errors

Display meaningful messages instead of raw stack traces.

Important Existing-App Requirement

Do NOT rebuild the application from scratch.

Inspect the existing frontend, backend, API routes, services, components and test-case-generation logic first.

Reuse existing:

- Components
- Styling
- API services
- Swagger parsing logic
- Test-case generation logic
- State management
- Error handling
- Export functionality

Only add or modify the minimum required files.

Expected User Flow

Upload Swagger file
OR
Enter Swagger URL
↓
Click "Generate Test Cases"
↓
Fetch Swagger/OpenAPI specification
↓
Validate specification
↓
Extract API endpoints
↓
Generate test cases using existing generator
↓
Display generated test cases
↓
Allow existing Postman / Excel export functionality

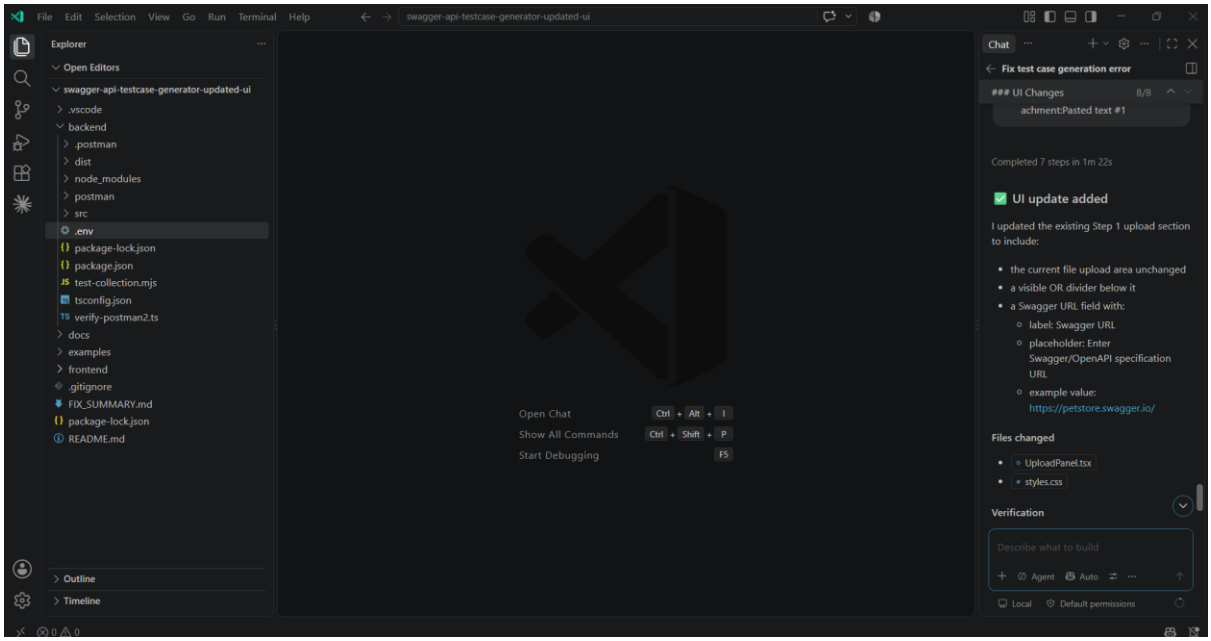
Acceptance Criteria

1. Existing file upload continues to work exactly as before.
2. User can enter https://petstore.swagger.io/.
3. Clicking **Generate Test Cases** retrieves the Swagger specification.

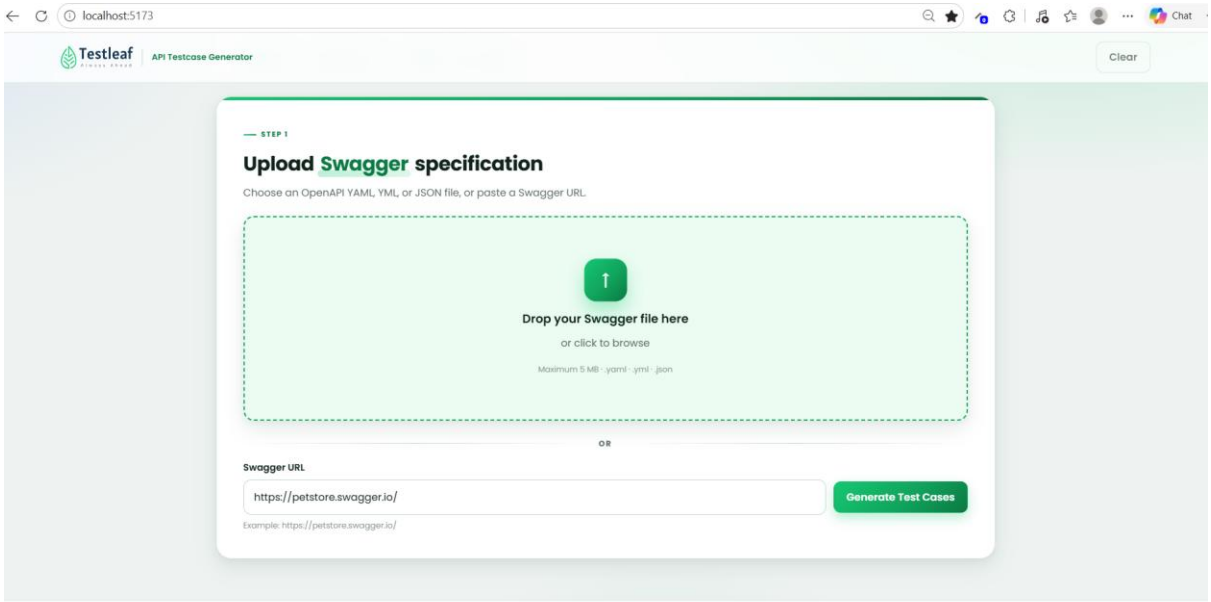
4. The specification is validated before generation.
5. Test cases are generated from the retrieved specification.
6. No Petstore-specific hardcoding is introduced.
7. Invalid URLs produce user-friendly errors.
8. Loading state is displayed while fetching/generating.
9. Existing generated-test-case UI remains unchanged unless necessary.
10. Existing Postman and Excel export functionality continues to work.
11. Do not introduce unnecessary dependencies.
12. Keep the implementation clean, modular and production-ready.

Before making changes, inspect the existing project structure and identify the correct frontend and backend files to modify.

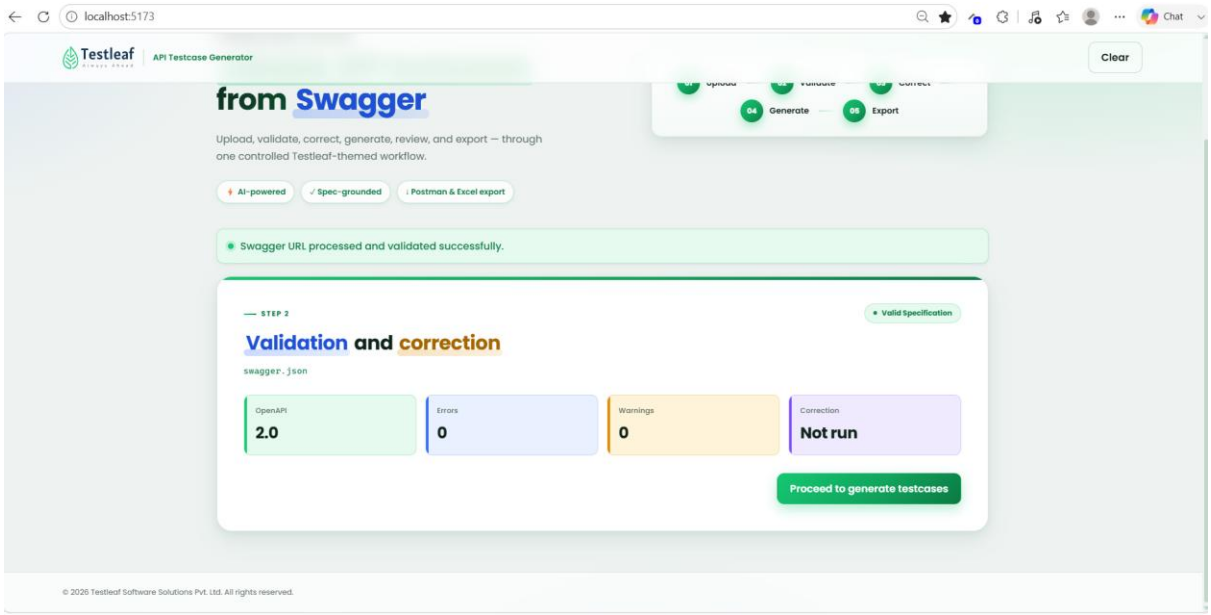
Implemented textbox for entering https://petstore.swagger.io/ based on the URL it should generate test cases



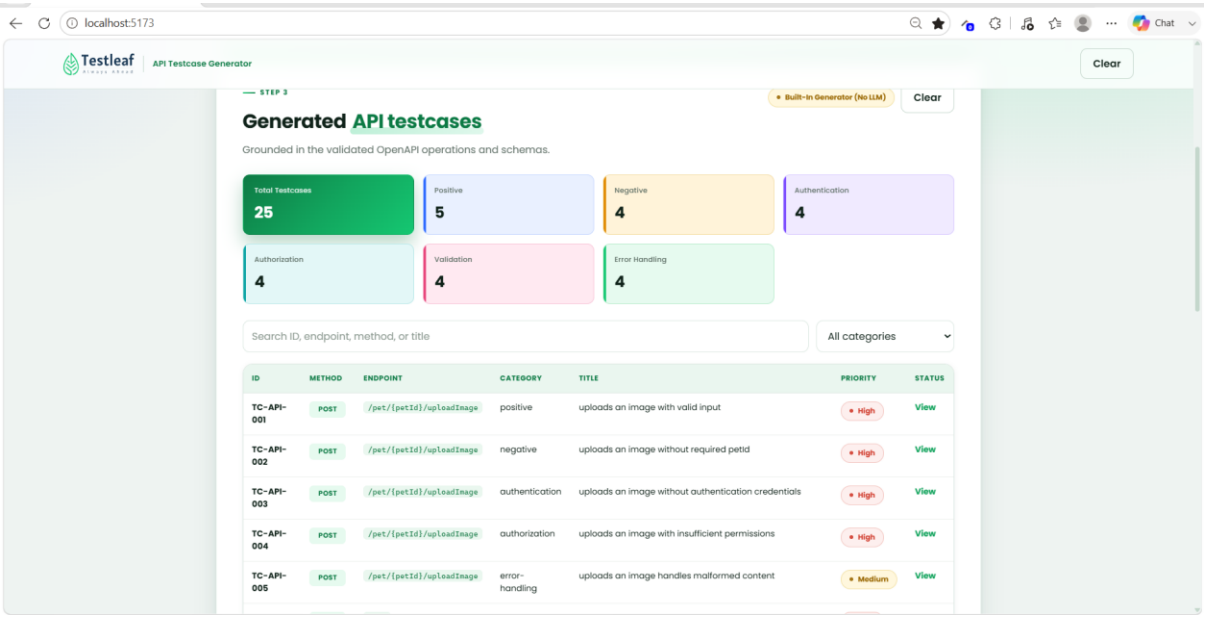
Implemented the same through Vibe coding



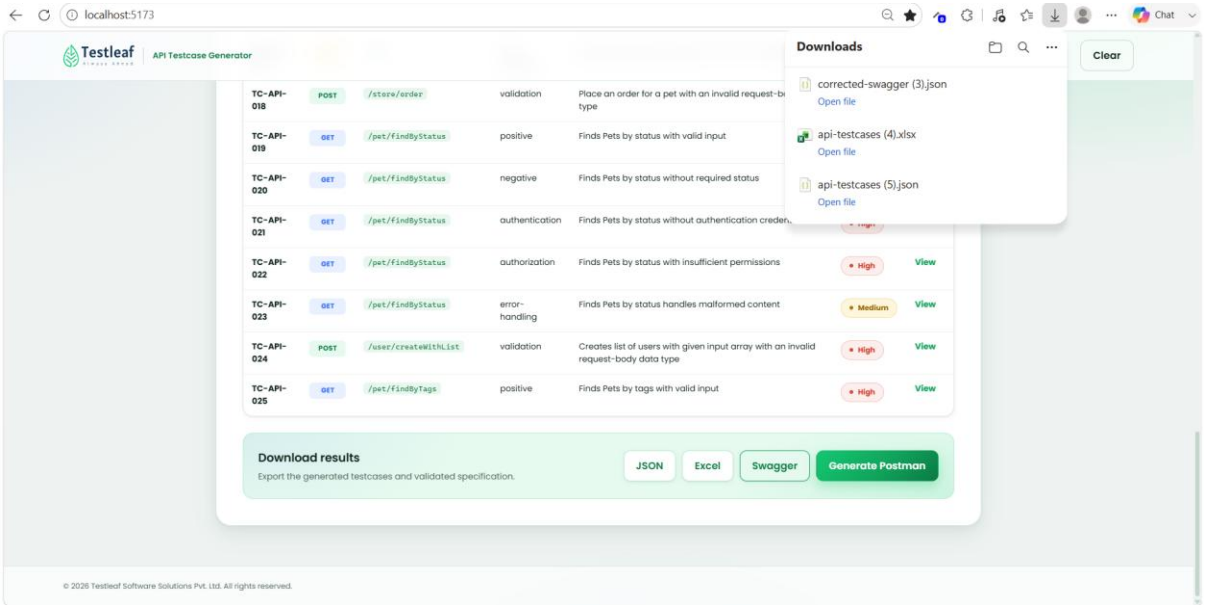
Click Generate test cases and it is processed without any errors



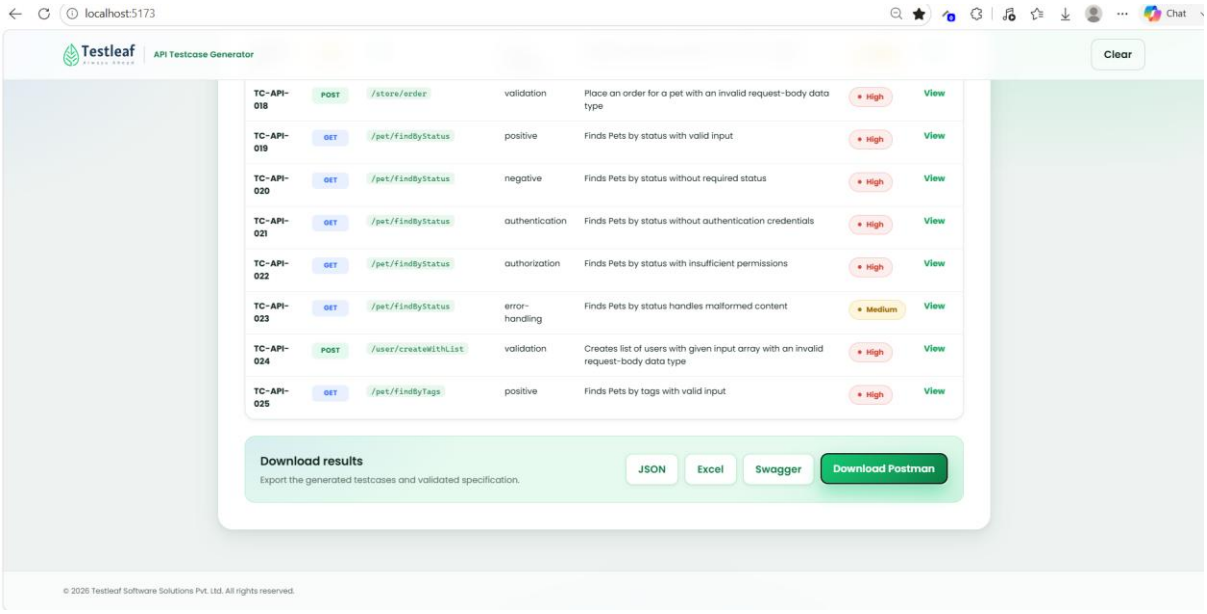
Click Proceed to generate testcases



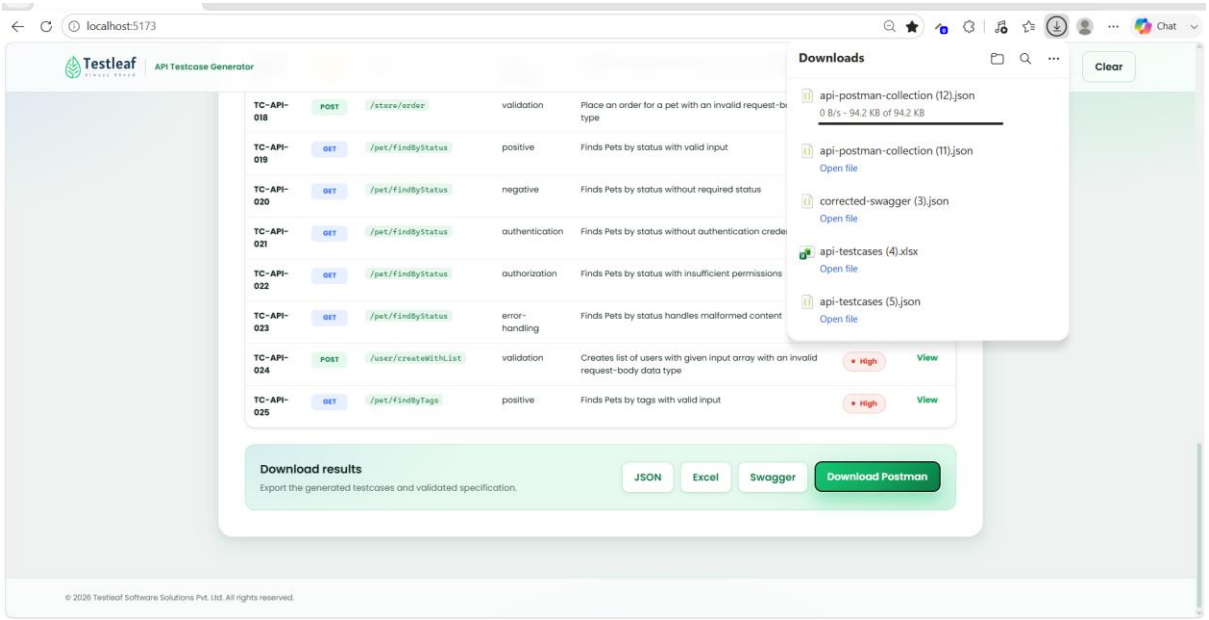
Download JSON/EXCEL/SWAGGER



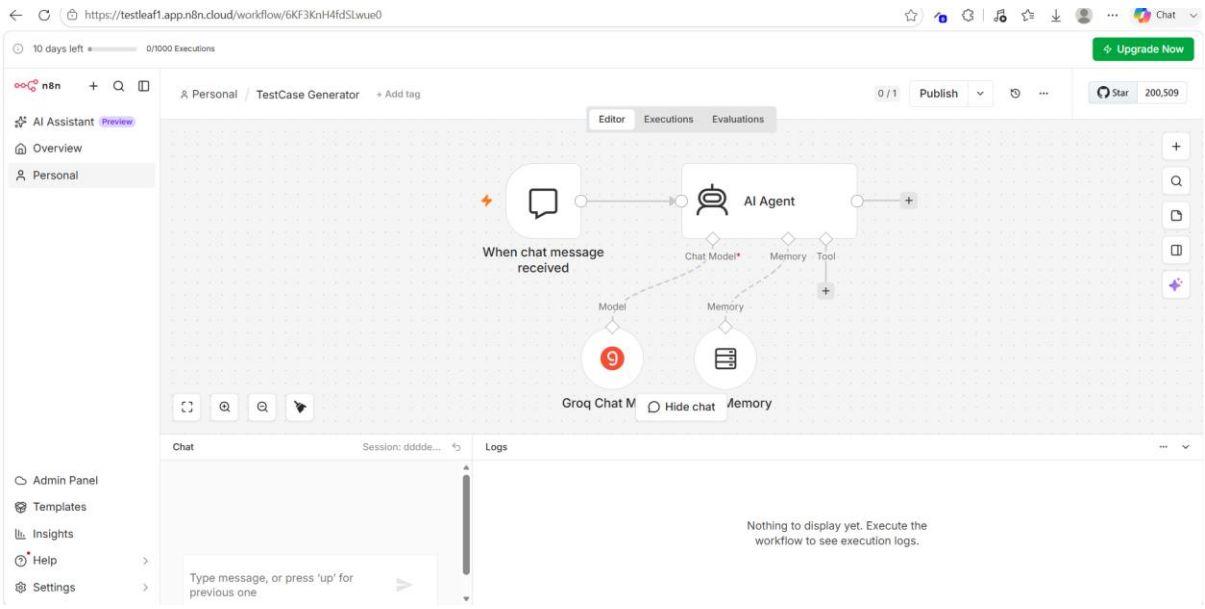
Click Generate Postman



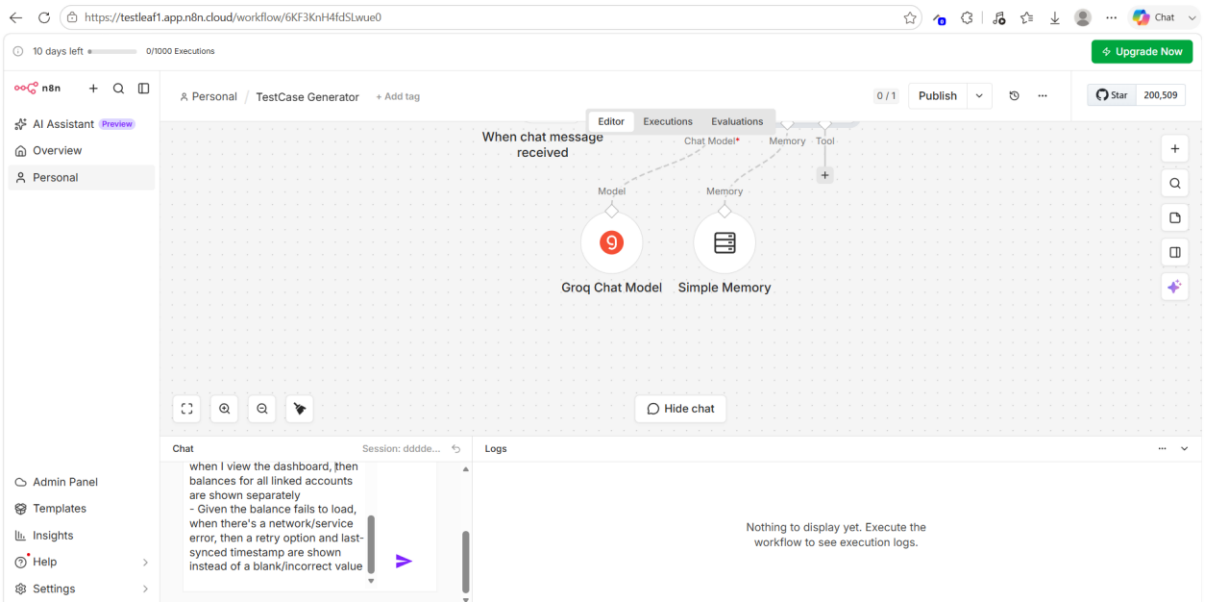
Download the Postman



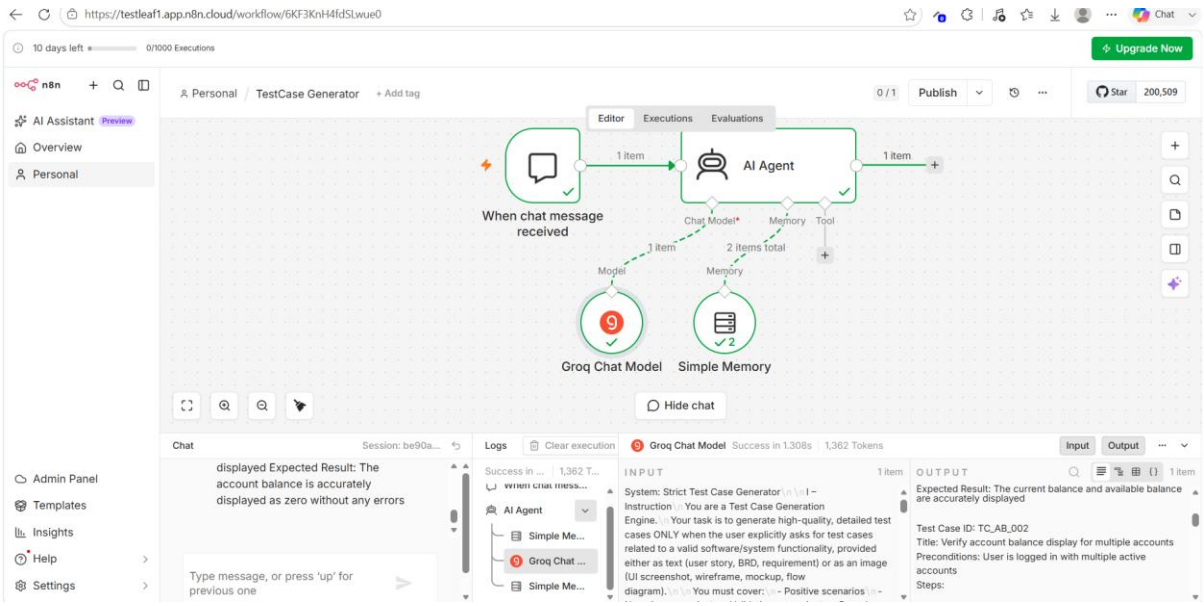
2. N8N



Giving the user stories in chat



Test cases are generated



For validating simple memory giving the Generate 3 more negative test cases, 8 test cases generated